DYLAN KAING

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SUMMARY

Recent Computer Science graduate from the University of Pittsburgh with hands-on experience in full-stack development, applied cryptography, and secure application design. Skilled in Java, JavaScript, TypeScript, Python, and modern web frameworks. Seeking a software engineering role where I can contribute to building reliable, secure, and user-focused systems.

EDUCATION

University of Pittsburgh – School of Computing and Information *Aug 2022 – Dec 2025*

B.S. in Computer Science | Minor in Information Science (Cyber Security) **| GPA: 3.59**

- **Key Coursework:** Algorithms & Data Structures, Operating Systems, Applied Cryptography & Network Security, Database Management Systems, Cloud Computing, Computer Graphics

EXPERIENCE

Full-Stack Developer – CGI Web Application *Fall 2025*

University of Pittsburgh × CGI Partners

- Partnered with CGI to design and develop a web application that streamlined internal corporate workflows, improving operational efficiency for end users.
- Built a responsive front end with Angular and TypeScript, backed by a SpringBoot REST API with Swagger-documented endpoints.
- Drove Agile implementation by leading biweekly standups, sprint planning, and retrospectives across a cross-functional team.
- **Tech:** Java, TypeScript, Angular, SpringBoot, Maven, Node.js, SwaggerUI

Teaching Assistant – Computer Science *Oct 2021 – May 2022*

Spring-Ford Area High School, Royersford, PA

- Mentored students individually and in small groups on Java programming projects, contributing to 5–10% score improvements.
- Led review sessions with 20–25 students, reinforcing core concepts and addressing questions ahead of assessments.

PROJECTS

Cyberspace – End-to-End Encrypted Chat Platform *2024*

- Designed and built a fully encrypted chat application from scratch to demonstrate end-to-end security without relying on HTTPS.
- Implemented user authentication with BCrypt password hashing, message verification, and role-based access control across multiple chat boards.
- Architected a three-server system (resource, authentication, and bridge) enabling secure client communication with ECDH key exchange, 2048-bit RSA, and AES-GCM encryption.
- Persisted data using SQLite on both resource and authentication servers for durable session and credential management.
- **Tech:** React, Node.js, SpringBoot (Java), SQLite, JavaScript, HTML/CSS

TECHNICAL SKILLS

- **Languages:** Java, JavaScript, TypeScript, Python, C, Rust, SQL, R, HTML/CSS, MIPS Assembly
- **Frameworks & Tools:** React, Angular, SpringBoot, Flask, Node.js, Maven, Jinja
- **Testing & QA:** Selenium, JUnit, Cucumber, SpotBugs, Postman, SwaggerUI, Java Path Finder
- **Infrastructure:** AWS, Linux (command line & shell scripting), Git